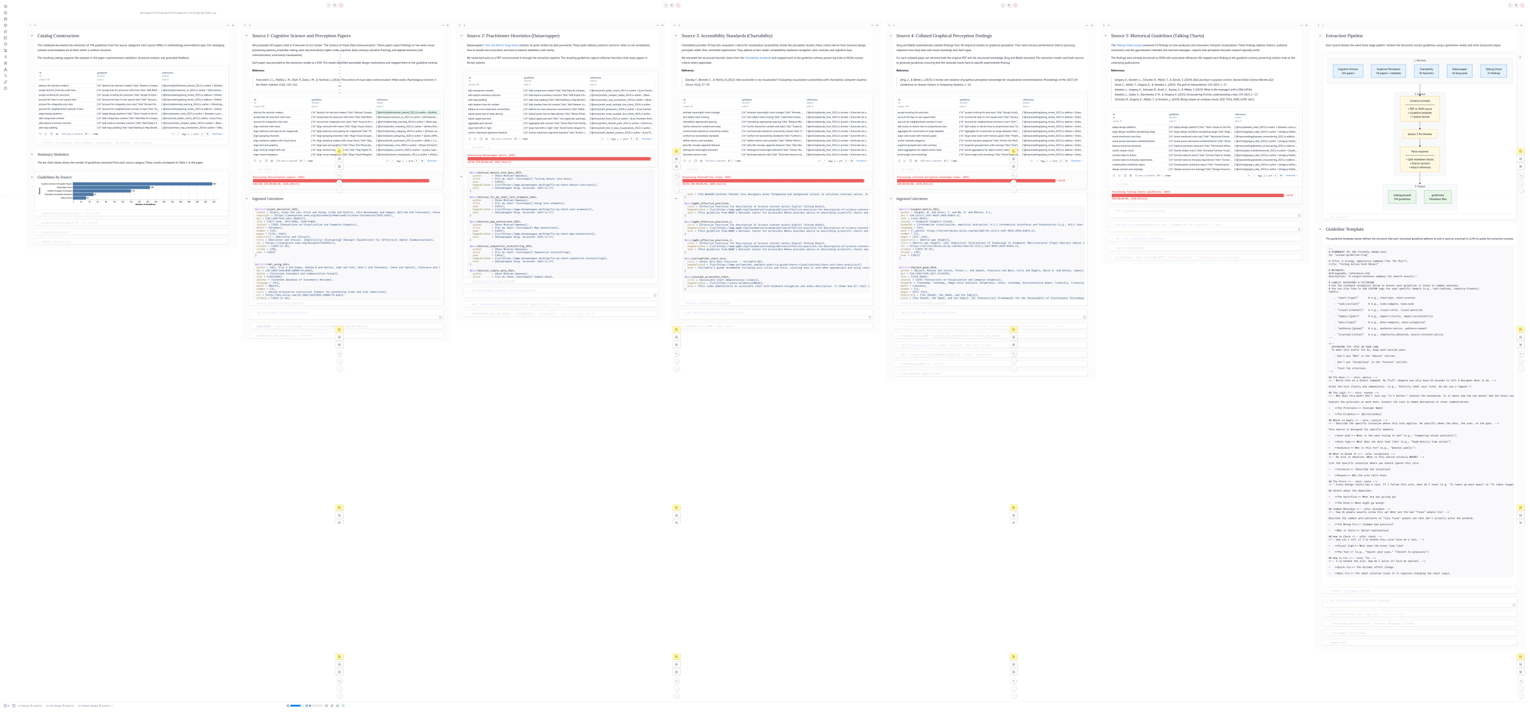

packages/catalog/python/examples/cataloging/index.py

## Catalog Construction

This notebook documents the extraction of 744 guidelines from five source categories. Each source differs in methodology and evidence type; the cataloging scheme accommodates all of them within a uniform structure.

The resulting catalog supports the analyses in the paper: expressiveness validation, structural analysis, and grounded feedback.

| id<br>str<br>unique: 744 | guideline<br>struct[7]<br>nulls: 0 | references<br>list[str]<br>nulls: 0 |
|---|---|---|
| abstract-for-decision-makers | {"id":"abstract-for-decision-makers","title":"Abstract Comple… | ["@article{kohlhammer_toward_2012,\n author = {Kohlha… |
| accept-bias-for-precision-with-ticks | {"id":"accept-bias-for-precision-with-ticks","title":"Add Refer… | ["@article{mccoleman_no_2021,\n author = {McColeman, … |
| accept-scrolling-for-precision | {"id":"accept-scrolling-for-precision","title":"Accept Scroll.in… | ["@inproceedings{zeng_review_2023,\n address = {Hamb… |
| account-for-bias-in-non-square-bars | {"id":"account-for-bias-in-non-square-bars","title":"Account … | ["@inproceedings{zeng_review_2023,\n address = {Hamb… |
| account-for-integrality-size-color | {"id":"account-for-integrality-size-color","title":"Account for … | ["@article{demiralp_learning_2014,\n author = {Demiralp,… |
| account-for-neighborhood-contrast-in-bars | {"id":"account-for-neighborhood-contrast-in-bars","title":"Ac… | ["@inproceedings{zeng_review_2023,\n address = {Hamb… |
| adapt-design-platform | {"id":"adapt-design-platform","title":"Tailor Visuals to the M… | ["@misc{koesten_what_2025,\n author = {Koesten, Laura … |
| add-comparison-context | {"id":"add-comparison-context","title":"Add Data for Compar… | ["@misc{muth_better_charts_2017,\n author = {Lisa Chart… |
| add-explicit-summary-columns | {"id":"add-explicit-summary-columns","title":"Add Explicit S… | ["@misc{mintzer_compact_tables_2024,\n author = {Rose … |
| add-map-padding | {"id":"add-map-padding","title":"Add Padding to Map Border… | ["@misc{mintzer_map_annotations_2024,\n author = {Ros… |

744 rows, 3 columns 10 / page Page 1 of 75 Download

```
catalog = tc_catalog + prc_catalog + ch_catalog + dw_catalog + misc_catalog
```

## Summary Statistics

The bar chart below shows the number of guidelines extracted from each source category. These counts correspond to Table 1 in the paper.

## Guidelines by Source

```
guidelines_by_source_chart = alt.layer(
```

```
guidelines_with_source_df = pl.concat(
```

```
catalogs_by_source = {
```

```
import altair as alt
```

## Source 1: Cognitive Science and Perception Papers

We processed 103 papers cited in Franconeri et al.'s review *"The Science of Visual Data Communication"*. These papers report findings on low-level visual processing (saliency, ensemble coding, color discrimination), higher-order cognition (bias, memory, narrative framing), and applied domains (risk communication, uncertainty visualization).

Each paper was provided to the extraction model as a PDF. The model identified actionable design implications and mapped them to the guideline schema.

**Reference:**

> *Franconeri, S. L., Padilla, L. M., Shah, P., Zacks, J. M., & Hullman, J. (2021). The science of visual data communication: What works. Psychological Science in the Public Interest, 22(3), 110–161.*

| id<br>str<br>unique: 338 | guideline<br>struct[7]<br>nulls: 0 | references<br>list[str]<br>nulls: 0 |
|---|---|---|
| abstract-for-decision-makers | {"id":"abstract-for-decision-makers","title":"Abstract Comple… | ["@article{kohlhammer_toward_2012,\n author = {Kohlha… |
| accept-bias-for-precision-with-ticks | {"id":"accept-bias-for-precision-with-ticks","title":"Add Refer… | ["@article{mccoleman_no_2021,\n author = {McColeman, … |
| account-for-integrality-size-color | {"id":"account-for-integrality-size-color","title":"Account for I… | ["@article{demiralp_learning_2014,\n author = {Demiralp,… |
| align-centroid-with-mean | {"id":"align-centroid-with-mean","title":"Align Visual Centroi… | ["@article{ondov_revealing_2021,\n author = {Ondov, Bria… |
| align-darkness-and-opacity-for-magnitude | {"id":"align-darkness-and-opacity-for-magnitude","title":"M… | ["@article{schloss_mapping_2019,\n author = {Schloss, Ka… |
| align-grouping-channels | {"id":"align-grouping-channels","title":"Align Visual Groupin… | ["@article{zacks_designing_2020,\n author = {Zacks, Jeffre… |
| align-sentence-subject-with-visual-focus | {"id":"align-sentence-subject-with-visual-focus","title":"Alig… | ["@article{roth_asymmetric_2012,\n author = {Roth, I. C. a… |
| align-text-and-graphics | {"id":"align-text-and-graphics","title":"Place Text Physically … | ["@article{mayer_nine_2003,\n author = {Mayer, Richard E… |
| align-vertical-height-with-risk | {"id":"align-vertical-height-with-risk","title":"Map Higher Ri… | ["@article{lipkus_numeric_2007,\n author = {Lipkus, Isaac … |
| align-visual-metaphors | {"id":"align-visual-metaphors","title":"Align Visual Metaphor… | ["@article{hullman_visualization_2011,\n author = {Hullm… |

338 rows, 3 columns 10 / page Page 1 of 34 Download

```
misc_tasks = [
```

```
Processing Miscellaneous papers: 100%|
103/103 [00:00<00:00, 3328.20it/s]
```

### Ingested Literature

```
@article{ajani_declutter_2022,
 author = {Ajani, Kiran and Lee, Elsie and Xiong, Cindy and Knaflic, Cole Nussbaumer and Kemper, William and Franconeri, Steve
 copyright = {https://ieeexplore.ieee.org/Xplorehelp/downloads/license-information/IEEE.html},
 doi = {10.1109/TVCG.2021.3068337},
 issn = {1077-2626, 1941-0506, 2160-9306},
 journal = {IEEE Transactions on Visualization and Computer Graphics},
 month = {October},
 number = {10},
 pages = {3351--3364},
 shorttitle = {Declutter and {Focus}},
 title = {Declutter and {Focus}: {Empirically} {Evaluating} {Design} {Guidelines} for {Effective} {Data} {Communication}},
 url = {https://ieeexplore.ieee.org/document/9385921/},
 urldate = {2025-11-18},
 volume = {28},
 year = {2022}
}


@article{akl_using_2011,
 author = {Akl, Elie A and Oxman, Andrew D and Herrin, Jeph and Vist, Gunn E and Terrenato, Irene and Sperati, Francesca and C
 doi = {10.1002/14651858.CD006776.pub2},
 editor = {{Cochrane Consumers and Communication Group}},
 issn = {14651858},
 journal = {Cochrane Database of Systematic Reviews},
 language = {en},
 month = {March},
 number = {3},
 title = {Using alternative statistical formats for presenting risks and risk reductions},
 url = {http://doi.wiley.com/10.1002/14651858.CD006776.pub2},
 urldate = {2025-11-18},
```

```
display_references(misc_catalog)
```

```
def misc_paper_to_catalog_entries(
```

```
misc_items = zot.everything(zot.collection_items(ZOTERO_MISC_COLLECTION_ID))
```

```
ZOTERO_MISC_COLLECTION_ID = "LXDQ9T85"
```

## Source 2: Practitioner Heuristics (Datawrapper)

Datawrapper's "Dos and Don'ts" blog series contains 36 posts written by data journalists. These posts address practical concerns: when to use annotations, how to handle axis truncation, and how to balance aesthetics with clarity.

We saved each post as a PDF and processed it through the extraction pipeline. The resulting guidelines capture editorial heuristics that rarely appear in formal systems.

| id<br>str<br>unique: 186 | guideline<br>struct[7]<br>nulls: 0 | references<br>list[str]<br>nulls: 0 |
|---|---|---|
| add-comparison-context | {"id":"add-comparison-context","title":"Add Data for Compari… | ["@misc{muth_better_charts_2017,\n author = {Lisa Charlot… |
| add-explicit-summary-columns | {"id":"add-explicit-summary-columns","title":"Add Explicit Su… | ["@misc{mintzer_compact_tables_2024,\n author = {Rose … |
| add-map-padding | {"id":"add-map-padding","title":"Add Padding to Map Borders"… | ["@misc{mintzer_map_annotations_2024,\n author = {Rose … |
| add-shadow-lines-for-context | {"id":"add-shadow-lines-for-context","title":"Add Shadow Lin… | ["@misc{muth_small_multiple_line_charts_2024,\n author … |
| adhere-to-cross-newsroom-conventions | {"id":"adhere-to-cross-newsroom-conventions","title":"Adher… | ["@misc{muth_partycolors_2018,\n author = {Lisa Charlott… |
| adjust-panel-size-to-data-density | {"id":"adjust-panel-size-to-data-density","title":"Resize Panel… | ["@misc{muth_small_multiple_line_charts_2024,\n author … |
| adjust-uppercase-text | {"id":"adjust-uppercase-text","title":"Use uppercase sparingly… | ["@misc{muth_fonts_2022,\n author = {Lisa Charlotte Muth… |
| aggregate-poll-sources | {"id":"aggregate-poll-sources","title":"Show Data from Multip… | ["@misc{jockers_election_polls_2021,\n author = {Simon Jo… |
| align-text-left-or-right | {"id":"align-text-left-or-right","title":"Avoid Center-Aligned Te… | ["@misc{muth_text_in_data_visualizations_2022,\n author … |
| anchor-important-segments-baseline | {"id":"anchor-important-segments-baseline","title":"Anchor t… | ["@misc{muth_stacked_columns_2018,\n author = {Lisa Ch… |

186 rows, 3 columns 10 / page Page 1 of 19 Download

```
dw_tasks = [
```

```
Processing Datawrapper posts: 100%|
36/36 [00:00<00:00, 2832.39it/s]
```

```
@misc{mintzer_donuts_into_bars_2025,
  author       = {Rose Mintzer-Sweeney},
  title        = {Fix my chart \textquotedbl{} Turning donuts into bars},
  year         = {2025},
  howpublished = {\url{https://www.datawrapper.de/blog/fix-my-chart-donuts-into-bars}},
  note         = {Datawrapper Blog. Accessed: 2025-11-17}
}

@misc{mintzer_fix_my_chart_text_elements_2024,
  author       = {Rose Mintzer-Sweeney},
  title        = {Fix my chart \textquotedbl{} Using text elements},
  year         = {2024},
  howpublished = {\url{https://www.datawrapper.de/blog/fix-my-chart-text-elements}},
  note         = {Datawrapper Blog. Accessed: 2025-11-17}
}

@misc{mintzer_map_annotations_2024,
  author       = {Rose Mintzer-Sweeney},
  title        = {Fix my chart \textquotedbl{} Map annotations},
  year         = {2024},
  howpublished = {\url{https://www.datawrapper.de/blog/fix-my-chart-map-annotations}},
  note         = {Datawrapper Blog. Accessed: 2025-11-17}
}

@misc{mintzer_sequential_storytelling_2024,
  author       = {Rose Mintzer-Sweeney},
  title        = {Fix my chart \textquotedbl{} Sequential storytelling},
  year         = {2024},
  howpublished = {\url{https://www.datawrapper.de/blog/fix-my-chart-sequential-storytelling}},
  note         = {Datawrapper Blog. Accessed: 2025-11-17}
}

@misc{mintzer_simple_data_2024,
  author       = {Rose Mintzer-Sweeney},
  title        = {Fix my chart \textquotedbl{} Simple data},
```

```
display_references(dw_catalog)
```

```
def datawrapper_post_pdf_to_catalog_entries(
```

```
def _datawrapper_path_to_post_url(path: pathlib.Path) -> str:
```

```
DATAWRAPPER_POST_PDF_PATHS = list((NB_ROOT / "datawrapper" / "posts").glob("*.pdf"))
```

## Source 3: Accessibility Standards (Chartability)

Chartability provides 50 heuristic evaluation criteria for visualization accessibility. Unlike the perception studies, these criteria derive from inclusive design principles rather than controlled experiments. They address screen reader compatibility, keyboard navigation, color contrast, and cognitive load.

We extracted the structured heuristic items from the Chartability workbook and mapped each to the guideline schema, preserving links to WCAG success criteria where applicable.

**Reference:**

> *Elavsky, F., Bennett, C., & Moritz, D. (2022). How accessible is my visualization? Evaluating visualization accessibility with Chartability. Computer Graphics Forum, 41(3), 57–70.*

| id<br>str<br>unique: 50 | guideline<br>struct[7]<br>nulls: 0 | references<br>list[str]<br>nulls: 0 |
|---|---|---|
| animate-meaningful-chart-changes | {"id":"animate-meaningful-chart-changes","title":"Animate … | ["@article{elavsky_how_2022,\n annote = {Concrete set o… |
| axis-labels-clear-missing | {"id":"axis-labels-clear-missing","title":"Label Axes Clearly … | ["@article{elavsky_how_2022,\n annote = {Concrete set o… |
| chartability-appropriate-spacing | {"id":"chartability-appropriate-spacing","title":"Balance Whi… | ["@article{elavsky_how_2022,\n annote = {Concrete set o… |
| clarify-interactive-context-and-state | {"id":"clarify-interactive-context-and-state","title":"Explicit… | ["@article{elavsky_how_2022,\n annote = {Concrete set o… |
| communicate-statistical-uncertainty-clearly | {"id":"communicate-statistical-uncertainty-clearly","title":"C… | ["@article{elavsky_how_2022,\n annote = {Concrete set o… |
| conform-to-accessibility-standards | {"id":"conform-to-accessibility-standards","title":"Conform t… | ["@article{elavsky_how_2022,\n annote = {Concrete set o… |
| define-metrics-and-variables | {"id":"define-metrics-and-variables","title":"Define Metrics … | ["@article{elavsky_how_2022,\n annote = {Concrete set o… |
| describe-visually-apparent-features | {"id":"describe-visually-apparent-features","title":"Describe… | ["@article{elavsky_how_2022,\n annote = {Concrete set o… |
| distinguish-meaningful-elements | {"id":"distinguish-meaningful-elements","title":"Ensure Me… | ["@article{elavsky_how_2022,\n annote = {Concrete set o… |
| eliminate-seizure-risks | {"id":"eliminate-seizure-risks","title":"Eliminate Seizure-Ind… | ["@article{elavsky_how_2022,\n annote = {Concrete set o… |

50 rows, 3 columns 10 / page Page 1 of 5 Download

```
ch_tasks = [
```

```
Processing Chartability items: 100%|
50/50 [00:00<00:00, 1880.15it/s]
```

```
  note = {The WebAIM Contrast Checker lets designers enter foreground and background colours to calculate contrast ratios. It
}

@misc{wgbh_effective_practices,
  title = {Effective Practices for Description of Science Content within Digital Talking Books},
  howpublished = {\url{https://www.wgbh.org/foundation/ncam/guidelines/effective-practices-for-description-of-science-content-
  note = {This guideline from WGBH's National Center for Accessible Media provides advice on describing scientific charts and
}

@misc{wgbh_effective_practices_2,
  title = {Effective Practices for Description of Science Content within Digital Talking Books},
  howpublished = {\url{https://www.wgbh.org/foundation/ncam/guidelines/effective-practices-for-description-of-science-content-
  note = {This guideline from WGBH's National Center for Accessible Media provides advice on describing scientific charts and
}

@misc{wgbh_effective_practices_3,
  title = {Effective Practices for Description of Science Content within Digital Talking Books},
  howpublished = {\url{https://www.wgbh.org/foundation/ncam/guidelines/effective-practices-for-description-of-science-content-
  note = {This guideline from WGBH's National Center for Accessible Media provides advice on describing scientific charts and
}

@misc{yellowfinbi_chart_axis,
  title = {Chart Axis Best Practices -- Yellowfin-BI},
  howpublished = {\url{https://www.yellowfinbi.com/best-practice-guide/charts-visualizations/chart-axis-best-practices}},
  note = {Yellowfin's guide recommends including axis titles and units, starting axes at zero when appropriate and using consi
}

@misc{youtube_accessible_chart,
  title = {Accessible Chart Demonstration (video)},
  howpublished = {\url{https://youtu.be/w6nta1B6MLQ}},
  note = {This video demonstrates an accessible chart with keyboard navigation and audio description. It shows how all chart i
}
```

```
display_references(ch_catalog)
```

```
def chartability_item_to_catalog_entry(
```

```
CHARTABILITY_ITEMS = json.loads(
```

## Source 4: Collated Graphical Perception Findings

Zeng and Battle systematically collated findings from 58 empirical studies on graphical perception. Their work extracts performance metrics (accuracy, response time, bias) that rank visual encodings and chart types.

For each collated paper, we retrieved both the original PDF and the structured knowledge Zeng and Battle extracted. The extraction model used both sources to generate guidelines, ensuring that the rationale traces back to specific experimental findings.

**Reference:**

> *Zeng, Z., & Battle, L. (2023). A review and collation of graphical perception knowledge for visualization recommendation. Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems, 1–16.*

| id<br>str<br>unique: 140 | guideline<br>struct[7]<br>nulls: 0 | references<br>list[str]<br>nulls: 0 |
|---|---|---|
| accept-scrolling-for-precision | {"id":"accept-scrolling-for-precision","title":"Accept Scrolli… | ["@inproceedings{zeng_review_2023,\n address = {Ham… |
| account-for-bias-in-non-square-bars | {"id":"account-for-bias-in-non-square-bars","title":"Accoun… | ["@inproceedings{zeng_review_2023,\n address = {Ham… |
| account-for-neighborhood-contrast-in-bars | {"id":"account-for-neighborhood-contrast-in-bars","title":"… | ["@inproceedings{zeng_review_2023,\n address = {Ham… |
| add-scales-or-decile-ticks-to-proportional-bars | {"id":"add-scales-or-decile-ticks-to-proportional-bars","tit… | ["@inproceedings{zeng_review_2023,\n address = {Ham… |
| aggregate-for-numerosity-on-large-datasets | {"id":"aggregate-for-numerosity-on-large-datasets","title":… | ["@inproceedings{zeng_review_2023,\n address = {Ham… |
| align-color-scale-with-memory-goals | {"id":"align-color-scale-with-memory-goals","title":"Match… | ["@inproceedings{zeng_review_2023,\n address = {Ham… |
| anchor-stacked-categories | {"id":"anchor-stacked-categories","title":"Anchor the Most … | ["@inproceedings{zeng_review_2023,\n address = {Ham… |
| augment-grouped-bars-with-overlays | {"id":"augment-grouped-bars-with-overlays","title":"Overl… | ["@inproceedings{zeng_review_2023,\n address = {Ham… |
| avoid-aggregation-for-object-centric-tasks | {"id":"avoid-aggregation-for-object-centric-tasks","title":"A… | ["@inproceedings{zeng_review_2023,\n address = {Ham… |
| avoid-angle-only-encodings | {"id":"avoid-angle-only-encodings","title":"Avoid Angle-On… | ["@inproceedings{zeng_review_2023,\n address = {Ham… |

140 rows, 3 columns 10 / page Page 1 of 14 Download

```
prc_tasks = [
```

```
Processing collated perception knowledge items: 100%|                | 58/58
[00:00<00:00, 2170.35it/s]
```

### Ingested Literature

```
@article{aigner_bertin_2011,
 author = {Aigner, W. and Kainz, C. and Ma, R. and Miksch, S.},
 doi = {10.1111/j.1467-8659.2010.01845.x},
 issn = {1467-8659},
 journal = {Computer Graphics Forum},
 keywords = {information visualization, empirical evaluation, H.5.2 Information Interfaces and Presentation (e.g., HCI): User
 language = {en},
 note = {\_eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1111/j.1467-8659.2010.01845.x},
 number = {1},
 pages = {215--228},
 shorttitle = {Bertin was {Right}},
 title = {Bertin was {Right}: {An} {Empirical} {Evaluation} of {Indexing} to {Compare} {Multivariate} {Time}-{Series} {Data} {
 url = {https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1467-8659.2010.01845.x},
 urldate = {2025-10-15},
 volume = {30},
 year = {2011}
}


@article{bujack_good_2018,
 author = {Bujack, Roxana and Turton, Terece L. and Samsel, Francesca and Ware, Colin and Rogers, David H. and Ahrens, James},
 doi = {10.1109/TVCG.2017.2743978},
 issn = {1941-0506},
 journal = {IEEE Transactions on Visualization and Computer Graphics},
 keywords = {taxonomy, Taxonomy, Image color analysis, Brightness, Color, colormap, discriminative power, linearity, Linearity
 month = {January},
 number = {1},
 pages = {923--933},
 shorttitle = {The {Good}, the {Bad}, and the {Ugly}},
 title = {The {Good}, the {Bad}, and the {Ugly}: {A} {Theoretical} {Framework} for the {Assessment} of {Continuous} {Colormaps
```

```
display_references(prc_catalog)
```

```
def collated_item_to_catalog_entries(
```

```
# A Review and Collation of Graphical Perception Knowledge for Visualization Recommendation (Zeng et al., 2023)
```

```
processed_zot_items = [process_zot_item(item) for item in zot_items]
```

```
def find_item_by_doi(all_items: list[dict], doi: str) -> dict:
```

```
zot = zotero.Zotero(VISFEEDBACK_GROUP_ID, "group", local=True)
```

```
VISFEEDBACK_GROUP_ID = 6228570
```

```
from pyzotero import zotero
```

## Source 5: Rhetorical Guidelines (Talking Charts)

The Talking Charts project produced 32 findings on how producers and consumers interpret visualizations. These findings address rhetoric, audience connection, and the gap between intended and received messages—aspects that perception-focused research typically omits.

The findings were already structured as JSON with associated references. We mapped each finding to the guideline schema, preserving citation links to the underlying publications.

**References:**

> *Gregory, K., Koesten, L., Schuster, R., Möller, T., & Davies, S. (2024). Data journeys in popular science. Harvard Data Science Review, 6(2).*
> *Knoll, C., Möller, T., Gregory, K., & Koesten, L. (2025). The gulf of interpretation. CHI 2025, 1–17.*
> *Koesten, L., Gregory, K., Schuster, R., Knoll, C., Davies, S., & Möller, T. (2025). What is the message? arXiv:2504.10544.*
> *Koesten, L., Šaske, A., Starchenko, S. M., & Gregory, K. (2025). Encountering friction, understanding crises. CHI 2025, 1–15.*
> *Schuster, R., Gregory, K., Möller, T., & Koesten, L. (2024). Being simple on complex issues. IEEE TVCG, 30(9), 6598–6611.*

| id<br>str<br>unique: 32 | guideline<br>struct[7]<br>nulls: 0 | references<br>list[str]<br>nulls: 0 |
|---|---|---|
| adapt-design-platform | {"id":"adapt-design-platform","title":"Tailor Visuals to the Me… | ["@misc{koesten_what_2025,\n author = {Koesten, Laura a… |
| align-design-workflow-storytelling-stage | {"id":"align-design-workflow-storytelling-stage","title":"Align… | ["@article{gregory_data_2024,\n author = {Gregory, Kathle… |
| avoid-emotional-color-bias | {"id":"avoid-emotional-color-bias","title":"Neutralize Emotio… | ["@inproceedings{koesten_encountering_2025,\n address … |
| avoid-purely-decorative-embellishments | {"id":"avoid-purely-decorative-embellishments","title":"Avoid… | ["@article{gregory_data_2024,\n author = {Gregory, Kathle… |
| balance-personal-relevance | {"id":"balance-personal-relevance","title":"Personalize Withe… | ["@inproceedings{koesten_encountering_2025,\n address … |
| clarify-simple-charts | {"id":"clarify-simple-charts","title":"Annotate Familiar Visuali… | ["@misc{saske_multidimensional_2025,\n author = {Saske,… |
| connect-data-to-action | {"id":"connect-data-to-action","title":"Connect Data to Tangib… | ["@article{gregory_data_2024,\n author = {Gregory, Kathle… |
| connect-data-to-everyday-experiences | {"id":"connect-data-to-everyday-experiences","title":"Connec… | ["@inproceedings{koesten_encountering_2025,\n address … |
| contextualize-unfamiliar-topics | {"id":"contextualize-unfamiliar-topics","title":"Contextualize … | ["@article{gregory_data_2024,\n author = {Gregory, Kathle… |
| design-around-core-message | {"id":"design-around-core-message","title":"Design Around a… | ["@article{gregory_data_2024,\n author = {Gregory, Kathle… |

32 rows, 3 columns 10 / page Page 1 of 4 Download

```
tc_tasks = [
```

```
Processing Talking Charts guidelines: 100%|                | 32/32
[00:00<00:00, 2426.07it/s]
```

```
def tc_guideline_to_catalog_entry(
```

```
def build_tc_prompt(guideline: dict, allowed_ref_ids: list[str]) -> str:
```

```
def list_tc_guideline_references(
```

```
import bibtexparser
```

```
tc_findings = json.loads((NB_ROOT / "talking-charts" / "findings.json").read_text())
```

## Extraction Pipeline

Each source follows the same three-stage pattern: retrieve the document, extract guidelines using a generative model, and write structured output.

```
mo.mermaid("""
```

## Guideline Template

The guideline template below defines the structure that each extracted guideline adheres to and is used as a prompt to LLMs to guide the extraction process.

```
---
# PERMANENT ID: URL-friendly kebab-case.
id: "unique-guideline-slug"

# TITLE: A strong, imperative command (The "Do This").
title: "Strong Action Verb Object"

# METADATA:
bibliography: references.bib
description: "A single-sentence summary for search results."

# LABELS: DISCOVERY & FILTERING
# Use the standard categories below to ensure your guideline is found in common searches.
# You are also free to add CUSTOM tags for your specific domain (e.g., tool:tableau, industry:finance).
labels:

   - "chart:[type]"       # e.g., chart:bar, chart:scatter

   - "task:[action]"      # e.g., task:compare, task:rank

   - "visual:[channel]"   # e.g., visual:color, visual:position

   - "impact:[goal]"      # e.g., impact:clarity, impact:accessibility

   - "data:[type]"        # e.g., data:temporal, data:categorical

   - "audience:[group]"   # e.g., audience:novice, audience:expert

   - "[custom]:[value]"   # e.g., complexity:advanced, source:internal-policy
---

<!--
  AUTHORING TIP: STAY IN YOUR LANE
  To make this useful for AI, keep each section pure.

   - Don't put "Why" in the "Advice" section.

   - Don't put "Exceptions" in the "Context" section.

   - Trust the structure.
-->

## The Rule <!-- role: advice -->
<!-- Write this as a direct command. No fluff. Imagine you only have 10 seconds to tell a designer what to do. -->

State the rule clearly and immediately. (e.g., "Directly label your lines. Do not use a legend.")

## The Logic <!-- role: reason -->
<!-- Why does this work? Don't just say "it's better." Explain the mechanism. Is it about how the eye moves? How the brain co

Explain the principle at work here. Connect the rule to human perception or clear communication.

*   **The Principle:** [Concept Name]

*   **The Evidence:** [@citationKey]

## Where to Apply <!-- role: context -->
<!-- Describe the specific situation where this rule applies. Be specific about the data, the user, or the goal. -->

This advice is designed for specific moments.

*   **User Goal:** What is the user trying to see? (e.g., "Comparing values precisely")

*   **Data Type:** What does the data look like? (e.g., "High-density time series")

*   **Audience:** Who is this for? (e.g., "General public")

## When to Break It <!-- role: exceptions -->
<!-- No rule is absolute. When is this advice actually WRONG? -->

List the specific scenarios where you should ignore this rule.

*   **Scenario:** [Describe the situation]

*   **Reason:** Why the rule fails here.

## The Price <!-- role: costs -->
<!-- Every design choice has a cost. If I follow this rule, what do I lose? (e.g. "It takes up more space" or "It takes longer

Be honest about the downsides.

*   **The Sacrifice:** What are you giving up?

*   **The Risk:** What might go wrong?

## Common Mistakes <!-- role: mistakes -->
<!-- How do people usually screw this up? What are the bad "fixes" people try? -->

Describe the common anti-patterns or "lazy fixes" people use that don't actually solve the problem.

*   **The Wrong Fix:** [Common bad practice]

*   **Why it fails:** [Brief explanation]

## How to Check <!-- role: check -->
<!-- How can I tell if I've broken this rule? Give me a test. -->

*   **Visual Sign:** What does the error look like?

*   **The Test:** (e.g., "Squint your eyes," "Convert to grayscale")

## How to Fix <!-- role: fix -->
<!-- I've broken the rule. How do I solve it? Give me options. -->

*   **Quick Fix:** The minimal effort change.

*   **Best Fix:** The ideal solution (even if it requires changing the chart type).
```

```
client = init_genai_client(
```

```
def display_references(catalog: Catalog):
```

```
def fence(string: str, lang: str = "json") -> str:
```

```
from typing import Callable, Iterable, ParamSpec, TypeVar
```

```
from google import genai
```

```
import json
```